

The empirical recurrence for OEIS A302956, recovered from its tail

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Abstract

OEIS A302956 records “Empirical recurrence of order 73” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 113$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 75$. Its last coefficient is $c_{73} = 664879104 \neq 0$, so no further tail satisfies anything shorter; any order-73 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302956 is “Number of $5 \times n$ 0..1 arrays with every element equal to 1, 2, 3, 4 or 5 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

0, 111, 3815, 80589, 2104533, 58679318, 1596912288, 43234526916, ...

The entry states:

Empirical recurrence of order 73 (see link above)

As of the “Last modified” line on the live entry (Apr 16 2018 11:43:32, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 113$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 113$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 73: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 73.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 226$ exact terms from $n = 2$ on, it returns order 73 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{73} c_i a(n-i),$$

c_1	11	c_{20}	7678651604884	c_{39}	−27808940996908051	c_{58}	−135846038255996
c_2	251	c_{21}	37842095123092	c_{40}	4584709180186540	c_{59}	192354562763432
c_3	3378	c_{22}	96893691823173	c_{41}	54339508177494973	c_{60}	−116336545496928
c_4	37624	c_{23}	69977371718057	c_{42}	−71598076198013180	c_{61}	52125608288616
c_5	210575	c_{24}	−10330564924391	c_{43}	73078503578490574	c_{62}	15427922550512
c_6	568783	c_{25}	−218723287371161	c_{44}	−33321395173569433	c_{63}	−8743625524496
c_7	−1930698	c_{26}	−955053594009322	c_{45}	−24128781416151432	c_{64}	1397455278256
c_8	−47164220	c_{27}	−761288217040097	c_{46}	58701569383410673	c_{65}	−2736090041408
c_9	−288311560	c_{28}	−487410904112963	c_{47}	−68995594401910893	c_{66}	−747527101056
c_{10}	−672906031	c_{29}	−542245555386360	c_{48}	44944768054900788	c_{67}	581032047296
c_{11}	333194933	c_{30}	4619173028112749	c_{49}	−12835618717132398	c_{68}	109141421056
c_{12}	9188527528	c_{31}	2819961073670154	c_{50}	−11492469706821720	c_{69}	45546477568
c_{13}	42939787309	c_{32}	2567253458424812	c_{51}	19352989917031175	c_{70}	11389418496
c_{14}	84831075849	c_{33}	9661615064861008	c_{52}	−13917278415270382	c_{71}	−14765423616
c_{15}	−4458588711	c_{34}	−13734855595180592	c_{53}	5421190453463558	c_{72}	−1783185408
c_{16}	−471653234591	c_{35}	605597190032335	c_{54}	1168967752051109	c_{73}	664879104
c_{17}	−2056080076990	c_{36}	−2448961503378771	c_{55}	−2675131900025760		
c_{18}	−4348054661828	c_{37}	−34933732691639570	c_{56}	1926312484259766		
c_{19}	−1900593832401	c_{38}	37028537180115784	c_{57}	−749599171908864		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 75$, and it is the recurrence OEIS A302956 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{73} - c_1x^{72} - \dots - c_{73}$. Its constant term is $-c_{73} = -664879104$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence

is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 73 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 73, so $\deg q = 73$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 73, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 73, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 664879104.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-73 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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